

RAGHAVAN BALANATHAN

raghavanb2103@gmail.com | +1 (765) 734-7606 | /in/raghavanbala/ | West Lafayette, IN | Portfolio

EDUCATION

Purdue University

M.S. in Computer Science

Courses: Computer Networks, Databases, Statistical Machine Learning

West Lafayette, IN, USA

May 2027 (Expected)

National Institute of Technology

B.Tech. (Honors) in Computer Science and Engineering | GPA: 3.88/4.0 | Rank: 1/141

Tiruchirappalli, India

May 2025

TECHNICAL SKILLS

- **Languages:** Python, C++, C, SQL, JavaScript, HTML/CSS, React
- **Cloud & DevOps:** Google Cloud (GCP), Docker, Kubernetes, CI/CD, Git, FSDP
- **Databases:** SQL (PostgreSQL, MySQL), NoSQL (MongoDB)
- **ML:** PyTorch, TensorFlow, scikit-learn, NumPy, pandas, LLMs (Llama, T5), Transformers, Huggingface
- **Core Concepts:** Data Structures, Algorithms, Object-Oriented Design (OOD), Distributed Systems, Microservices, REST APIs

EXPERIENCE

Spring Lab-Indian Institute of Technology Madras

Research Intern

Chennai, India

Dec 2023 – Apr 2025

- Built and optimized a multi-modal ASR pipeline using self-supervised models (HuBERT) and LLMs, achieving less than 10% WER on a large-scale speech dataset.
- Designed and implemented a distributed training framework using Fully Sharded Data Parallel (FSDP) and QLoRA to fine-tune LLMs (Llama, GPT) on a 1 TB text and speech corpus, improving GPU memory efficiency and training throughput.
- Developed an end-to-end spoken keyword detection system using a custom transformer model, achieving 0.8 MTWV on 35.7K speech keywords, enabling scalable real-time audio search.

PROJECTS

Continual Learning for Video Segmentation of Surgical Tool

B.Tech Thesis

Tiruchirappalli, India

Jan 2025 – May 2025

- Developed a Domain-Incremental Continual Learning framework in PyTorch by integrating SAM2 with LoRA, enabling privacy-preserving model adaptation in surgical video streams.
- Designed a novel parameter selection algorithm using CLIP and K-Means, improving IoU metric by 5% over LwF baselines.

SummarizeIQ: Article Summarizer

Schlumberger's Hackathon, IIT Madras

Chennai, India

Jan 2023 – Feb 2023

- Designed a scalable microservice using FastAPI and TorchServe to deploy a T5 Transformer model for abstractive NLP summarization, reducing inference latency by 35%.
- Built asynchronous web scraping and batch processing pipelines integrated with LexRank/TextRank, improving throughput.
- Owned a full-stack, scalable solution, securing 4th place out of 1,150 teams.

PUBLICATIONS

- Adethya, G., Nitish, N., **Balanathan, R.**, Sitara, K. *Domain-Incremental Continual Learning for Robust Surgical Tool Segmentation*. MICCAI Workshop, 2025.